Ant-Inspired Visual Navigation Using Mushroom Body Neural Model in the CARLA Simulator

Sami Fergague
Master SIC – ETIS Laboratory, CNRS UMR 8051
CY Cergy Paris Université / ENSEA
Email: sami.fergague@ensea.fr

Nicolas Cuperlier
ETIS Laboratory, CNRS UMR 8051
CY Cergy Paris Université / ENSEA
Email: nicolas.cuperlier@cyu.fr

***Abstract*—This paper presents the implementation and evaluation of an ant-inspired visual navigation system for an autonomous vehicle in the CARLA urban driving simulator. Inspired by the neural mechanisms of desert ants (*Cataglyphis*), we implement a route-following pipeline based on the Mushroom Body (MB) neural model, which encodes visual scenes through low-resolution panoramic images processed via Sobel gradient extraction. The vehicle acquires a 360° fisheye view through a cubemap reprojection strategy, learns a route in a single pass (one-shot learning), and navigates autonomously using a differential familiarity signal from two parallel MB circuits. We evaluate the system through three experimental protocols: (i) repeated baseline navigation (80% success, median cross-track error 0.53 m, maximum 3.78 m), (ii) lateral offset robustness (attraction basin up to ± 2 m), and (iii) weather sensitivity (complete failure under rain, fog, sunset, and night). These results validate that lightweight insect-inspired navigation is viable in a realistic urban simulator, while revealing the fundamental limitation of purely appearance-specific visual encoding under dynamic illumination changes.**

***Index Terms*—Bio-inspired navigation, Mushroom body, Visual route following, Autonomous driving, One-shot learning, CARLA simulator, Neuromorphic model**

I. Introduction

Autonomous navigation in GPS-denied environments remains a major challenge in robotics, with applications ranging from space exploration to last-mile delivery [2], [3]. While modern approaches based on SLAM or deep learning achieve high accuracy, they typically require significant computational resources and large amounts of training data. In contrast, solitary foraging desert ants (*Cataglyphis velox*, *Melophorus bagoti*) navigate over hundreds of meters along familiar routes using minimal neural resources (a brain containing fewer than one million neurons) and a single learning trial, without relying on dead reckoning once routes are established [4], [16].

Ants rely primarily on visual cues processed through the Mushroom Bodies (MB)—paired neural structures involved in olfactory and visual learning [13]. These structures encode visual scenes as sparse representations through Kenyon Cells (KCs), enabling efficient familiarity-based route recognition [1]. Recently, Gattaux et al. demonstrated that this biological model can be transferred to a compact car-like robot (AntCar). Their route-centric lateralized MB architecture achieved continuous route-following with remarkable performance: a median lateral error of less than 25 cm over 1.6 km of real-world trials, using only 800-pixel input and 18.75 kB of memory per 50 m route [10].

However, transitioning such biomimetic models from small-scale robotic platforms to more complex environments—such as urban traffic scenarios—remains an open question. Driving simulators like CARLA [8] offer a controlled, reproducible testbed for evaluating navigation algorithms under realistic urban conditions without the constraints and risks of physical deployment.

Contributions. In this work, we extend this ant-inspired approach to an autonomous vehicle in an urban driving context using the CARLA simulator. Specifically, our contributions are:

1) A complete pipeline adapting the lateralized MB neural model to CARLA, including cubemap-based 360° panoramic acquisition mimicking the wide field-of-view of ant visual systems.
2) A quantitative baseline over repeated runs establishing performance metrics in an urban environment.
3) An analysis of the model's attraction basin via lateral offset experiments, demonstrating its capability to actively converge toward the learned route.
4) A study of weather sensitivity revealing the fundamental limitations of purely appearance-based navigation under dynamic illumination.

II. Related Work

A. Ant Visual Navigation

Desert ants navigate using a combination of path integration and visual route memories [7], [16]. The visual component relies on comparing the current panoramic view with stored memories to assess scene familiarity. Traditional models suggest that ants scan their environment by performing body rotations, selecting the direction of maximum familiarity—the visual compass [5], [22].

Recent neurobiological studies [1], [13] revealed that the Mushroom Body (MB) circuit compresses visual information into sparse, high-dimensional representations via the Kenyon Cells (KCs), with learning occurring at the KC-to-MBON (Mushroom Body Output Neuron) synapses through a single exposure (one-shot learning) [4].

B. Insect-Inspired Robot Navigation

Several works have transferred insect navigation principles to robots [3]. The AntCar system [9], [10] specifically addresses route-following using an MB neural model on a car-like robot equipped with a 220° fisheye camera. Key features include: one-shot learning, low-resolution input (down to 32×32 pixels [12]), a biologically constrained MB model with random projections, and a differential familiarity signal from two lateralized MB circuits for continuous steering control [10], [15].

A major advantage of the MB approach over conventional methods is its extremely low memory footprint: the entire route memory is stored in $2 \times N_{KC}$ synaptic weights (e.g., around 120 kB for $N_{KC} = 15{,}000$), regardless of the route length. In contrast, visual SLAM systems [2] require storing keyframes and point clouds that grow linearly with the explored area, and deep learning approaches require millions of parameters.

C. Visual Place Recognition

Visual Place Recognition (VPR) [14] addresses the problem of recognizing previously visited locations despite appearance changes. State-of-the-art methods like NetVLAD and SeqSLAM [17] use deep features or sequence matching to achieve robustness to illumination and viewpoint changes. However, these methods require orders of magnitude more computation and memory than the MB model. The trade-off between robustness and computational efficiency is central to our study: the MB model achieves reliable navigation under stable conditions with minimal resources, but sacrifices robustness to severe appearance changes.

D. CARLA Simulator

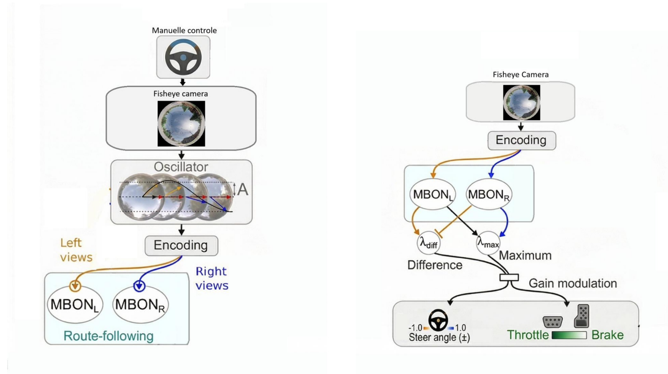
CARLA [8] is an open-source urban driving simulator providing realistic rendering, configurable weather conditions, and controllable vehicle dynamics. It supports flexible sensor configurations and has become a standard platform for evaluating autonomous driving systems, enabling systematic robustness studies (such as dynamic lighting or adverse weather) that would be highly impractical or time-consuming to conduct solely on physical robots.

III. Methods

A. System Overview

Our pipeline (Fig. 1) consists of five phases: (1) static route capture in CARLA; (2) data augmentation and MB training; (3) test route capture; (4) offline evaluation; (5) closed-loop autonomous navigation.

During the learning phase (Fig. 1a), the vehicle follows the route under static teleportation, and the MB circuits are trained on a single pass of augmented panoramic images. During navigation (Fig. 1b), the trained model computes a differential familiarity signal that drives real-time steering and speed commands.



(a) Learning phase (b) Navigation phase

Fig. 1: Overview of the proposed navigation pipeline. **(a)** During learning, panoramic images captured along the route are augmented by virtual angular oscillations and used to train two MB circuits (MB_L and MB_R) via one-shot anti-Hebbian learning. **(b)** During navigation, the current fisheye view is encoded and the differential familiarity $\Delta\lambda = \lambda_1 - \lambda_2$ between the two circuits drives the steering command, while $\lambda_{\min}$ modulates speed.

B. Panoramic Image Capture in CARLA

Since CARLA does not natively provide a 360° fisheye camera, we construct a synthetic one to emulate the wide field-of-view (FOV) of insect vision. A rig of six perspective cameras is arranged in a cube-map configuration on the simulated vehicle. Each camera captures a 512×512 pixel image with a 90° FOV, covering the front, back, left, right, up, and down directions. These six images are then projected into a single equidistant fisheye image of 512×512 pixels with a configurable maximum zenith angle $\theta_{\max}$ (set to $110°$ in our experiments).

The fisheye projection is computed using precomputed look-up tables that map each output pixel (u, v) in the fisheye image to the corresponding pixel in one of the six cube-map faces. For a given pixel at distance r from the image centre, the zenith angle is computed as:

$$\theta = \frac{r}{f_{\text{fish}}}, \quad f_{\text{fish}} = \frac{r_{\max}}{\theta_{\max}} \tag{1}$$

where $r_{\max}$ is half the image dimension. The 3D ray direction is then determined from θ and the azimuthal angle ϕ, and mapped to the appropriate cube face using dominant-axis selection.

C. Visual Encoding

Following the approach of Gattaux et al. [10], we apply a bio-inspired visual encoding pipeline to the captured fisheye images (Fig. 2). First, only the green channel is retained, mimicking the spectral sensitivity of ant photoreceptors [19]. A Gaussian filter ($\sigma = 3$ pixels) is then applied, corresponding to an acceptance angle of approximately $4.4°$ [20]. The image is downsampled to 32×32 pixels using nearest-neighbour interpolation, achieving a visual acuity of 0.145 pixel per degree,

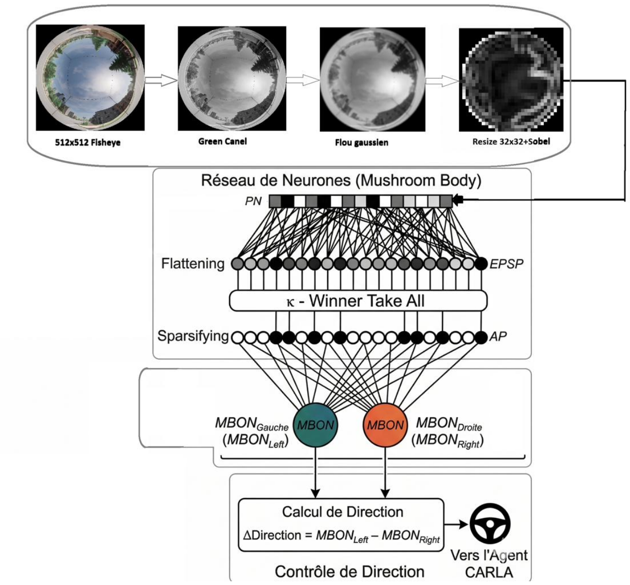


Fig. 2: Visual encoding and Mushroom Body architecture. Top: the 512×512 fisheye image is processed through green-channel extraction, Gaussian blurring, downsampling to 32×32, and Sobel edge detection to yield ∼800 Projection Neurons (PNs). Bottom: the PN vector is expanded to 15,000 Kenyon Cells via sparse random projections, sparsified by κ-Winner-Take-All, and projected onto two MBONs. The differential output $\Delta\text{Direction} = \text{MBON}_L - \text{MBON}_R$ drives the steering command.

consistent with the $\sim 7.1°$ inter-receptor angle measured in ants [20]. A Sobel filter then extracts edges, mimicking lateral inhibition in insect optical lobes [21]. Finally, the processed image is flattened into a vector of $n \approx 800$ Projection Neurons (PNs), retaining only pixels within a circular disk mask.

D. Mushroom Body Neural Network

The MB model follows the architecture described in [1], [10] and consists of three layers: Projection Neurons (PNs), Kenyon Cells (KCs), and Mushroom Body Output Neurons (MBONs) (Fig. 2).

1) PN to KC Expansion: The PN vector ($n \approx 800$ active pixels) is expanded into a high-dimensional KC representation through a fixed, sparse, pseudo-random binary synaptic matrix $\mathbf{W}_{\text{PNtoKC}} \in \{0,1\}^{N\times n}$, where each KC receives input from exactly $s = 4$ randomly selected PNs. The excitatory post-synaptic projection (EP) is computed as:

$$\text{EP} = \mathbf{W}_{\text{PNtoKC}}^{T} \cdot \text{PN} \tag{2}$$

A κ-Winner-Take-All mechanism selects the top κ% of KCs with the highest EP values, producing a sparse binary Action Potential (AP) vector:

$$\text{AP}_i = \begin{cases} 1, & \text{if } \text{EP}_i \in \{\kappa\text{-largest of EP}\} \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

We use $N_{\text{KC}} = 15{,}000$ and $\kappa = 0.005$, yielding $N_\kappa = 75$ active KCs per input (0.5% sparsity).

Adaptation of κ for CARLA. In the original AntCar robot [10], images are captured at 38 Hz (approximately 1 cm intervals), producing highly correlated successive frames. In CARLA, images are captured at 1 m intervals, making them significantly less correlated. With the original $\kappa = 0.01$ and 3,000 training patterns, KC saturation leaves only 2.5% of cells responsive. Reducing to $\kappa = 0.005$ preserves approximately 14.5% of active KCs, yielding a sufficient familiarity dynamic range ($\Delta u \approx 0.15$). This adaptation was validated empirically by monitoring KC saturation during training.

2) Learning Rule: Learning occurs through one-shot anti-Hebbian synaptic depression. The weight vector $\mathbf{W}_{\text{KCtoMBON}} \in \{0,1\}^N$ is initialized to all ones. For each training image, active KCs have their output synapse permanently depressed:

$$W_{\text{KCtoMBON},i} \leftarrow \begin{cases} 0, & \text{if } \text{AP}_i = 1 \\ W_{\text{KCtoMBON},i} & \text{otherwise} \end{cases} \tag{4}$$

The unfamiliarity of a view is then:

$$u = \frac{1}{N_\kappa} \sum_{i=1}^{N_{\text{KC}}} \text{AP}_i \cdot W_{\text{KCtoMBON},i} \tag{5}$$

where N_κ is the number of active KCs. A familiar view yields $u \approx 0$; a novel view yields $u \approx 1$. The familiarity index is $\lambda = 1 - u$.

3) Dual MB Architecture: Two independent MB circuits (MB1, MB2) with distinct random projections (seeds 42 and 99) are trained on the same route with opposite angular augmentations: MB1 with positive oscillations [0°, +5°, . . ., +45°] and MB2 with negative oscillations [0°, −5°, . . ., −45°], in steps of 5°. During navigation, the differential familiarity signal:

$$\Delta\lambda = \lambda_1 - \lambda_2 \tag{6}$$

indicates the direction of higher familiarity and drives the steering command, implementing the visual compass mechanism described in [10].

E. Training Procedure

During training (Phase 2), each route image is augmented by virtual angular oscillation: the panoramic image is rotated in the horizontal plane from −45° to +45° in steps of 5°, producing $2\times10 = 20$ augmented views per image (10 positive angles for MB1, 10 negative angles for MB2, including the 0° canonical view). This simulates the scanning behaviour of ants, who rotate their body to assess familiarity across directions.

Each MB is trained on $300 \times 10 = 3{,}000$ patterns via the anti-Hebbian rule, with the entire training completed in a single epoch—consistent with the biological one-shot learning constraint. The memory footprint of the trained model is $2 \times 15{,}000 \times 4 = 120$ kB (two weight vectors stored as `float32`), independent of route length.

F. Navigation Control

The steering command uses exponential smoothing to filter the noisy MB signal:

$$s_t = \alpha \cdot s_{t-1} + (1-\alpha) \cdot K_P \cdot \Delta\lambda_{\text{norm}} \quad (7)$$

with $\alpha = 0.15$, $K_P = 25$, clipped at ± 0.75. The normalised differential signal is $\Delta\lambda_{\text{norm}} = (u_1 - u_2)/(u_1 + u_2)$. The smoothing is critical: the raw MB signal oscillates in sign every ∼7 ticks due to KC quantization noise. With $\alpha = 0.15$, only the consistent directional signal emerges over ∼15 ticks.

The control loop runs at approximately 5 Hz (one image every 4 simulator ticks at 20 FPS), matching the effective decision rate of the MB signal. Speed is modulated by two factors, taking the minimum:

1) **Familiarity-based:** speed scales linearly with $\lambda_{\min} = \min(\lambda_1, \lambda_2)$ between $\lambda_{\text{slow}} = 0.92$ and $\lambda_{\text{fast}} = 0.996$.
2) **Steering-based:** speed decreases linearly when $|s_t| > 0.10$, reaching the minimum at $|s_t| \geq 0.40$.

Speed ranges from 8 km/h (straight line) to 2 km/h (tight curves). The target speed is converted to a throttle command via a linear mapping clipped to $[0.10, 1.0]$, ensuring a minimum traction force even at low speed. An emergency stop triggers if $\lambda_{\min} < 0.75$ for 40 consecutive ticks. Table IV summarises all key parameters.

IV. Experimental Setup

A. CARLA Configuration

Experiments were conducted in the CARLA simulator (version 0.9.15) running in synchronous mode at 20 FPS. We used the **Town05** map, which provides a diverse urban environment with buildings, vegetation, and road infrastructure.

The simulated vehicle (Tesla Model 3 blueprint) was equipped with a rig of six RGB cameras mounted at 2.0 m height to capture the cube-map panorama. The vehicle's physics were disabled during the learning phase (Phases 1–4) to allow precise teleportation along the route waypoints, and enabled during the navigation phase (Phase 5).

B. Route and Environment

The learning route spans 300 m along 300 waypoints (1 m spacing) in CARLA Town05 (Fig. 3). It includes a ∼90° right-hand curve and two straight segments, traversing an urban environment with buildings, vegetation, and road infrastructure. Starting from spawn point 152, waypoints are generated by following the road lane using CARLA's waypoint API.

C. Evaluation Metrics

We evaluate the model using two standard metrics [2]:

Cross-Track Error (XTE): the perpendicular distance from the vehicle's position to the nearest point on the learned route centreline. Following [10], we report the median ± Median Absolute Deviation (MAD) of per-run median values, and separately the maximum absolute XTE observed across all steps.

Fig. 3: The 300 m learning route in CARLA Town05, starting at the bottom-right (Départ) and ending at the top-left (Arrivée). The route includes a ∼90° right-hand curve followed by a straight segment.

Heading Error (HE): the angular difference between the vehicle's heading and the local route tangent. Reported as median ± MAD of per-run median values.

A run is considered *successful* if the vehicle reaches the last waypoint (WP 298) without triggering the emergency stop.

D. Experimental Protocols

1) Protocol 1: Baseline (10 runs): Ten identical runs with the same initial conditions: starting position aligned with the route, 0 m lateral offset, ClearNoon weather. This quantifies the stochastic variability of the closed-loop system.

2) Protocol 2: Lateral Offset (10 runs): One run per offset: −2.0, −1.5, −1.0, −0.5, 0.0, +0.5, +1.0, +1.5, +2.0 m, plus a second run at 0.0 m for reproducibility. ClearNoon weather. This characterises the model's attraction basin.

3) Protocol 3: Weather Sensitivity: We additionally conducted a qualitative investigation of the system's sensitivity to weather changes by testing the trained model (trained under ClearNoon) under four alternative conditions: Night, HardRain, Fog, and Sunset. Given the strongly adverse and expected results, this protocol was treated as a sanity check rather than a quantitative benchmark.

V. Results

A. Offline Evaluation (Phase 4)

Before testing closed-loop navigation, we validated the model's recognition capability through a static offline test. Each of the 250 test images (captured at 0 m lateral offset from the learned route) was presented to both MB circuits, and the unfamiliarity was recorded.

Fig. 4 shows the results. Mean unfamiliarity is $\bar{u}_{\text{MB1}} = 0.015$ and $\bar{u}_{\text{MB2}} = 0.013$, with 91% of steps below $u = 0.05$ for MB1. The occasional peaks correspond to locations where the visual scene changes abruptly (building corners, road junctions). These results confirm that the MB model can reliably recognise the route from a displaced position, validating the prerequisite for autonomous navigation.

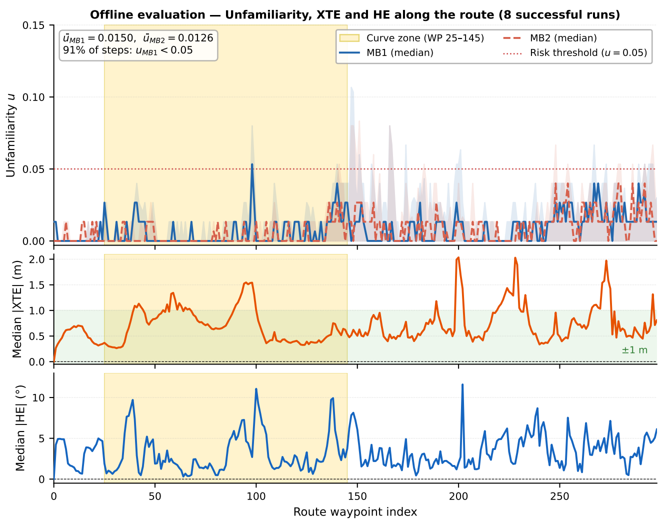


Fig. 4: Navigation evaluation along the 300 m route (8 successful runs, median aggregated per waypoint). **Top:** Unfamiliarity u for MB1 (blue) and MB2 (red dashed), with inter-run Q75 shaded envelope. Both circuits remain below the risk threshold ($u = 0.05$) for 91% of steps. Peaks correspond to visually salient transitions (corners, junctions). **Middle:** Median |XTE| stays within ± 1 m (green band) over most of the route, with higher values in the post-curve straight (WP 145–200). **Bottom:** Median |HE| remains below 5° outside the curve zone. Yellow band: curve zone (WP 25–145).

TABLE I: Baseline results (10 runs, ClearNoon, 0 m offset). Values are median $\pm$ MAD of per-run statistics, following [10].

Metric	**Successful** ($n = 8$)	**All** ($n = 10$)
Success rate	80%	—
Median |XTE| (m)	0.53 ± 0.06	0.60 ± 0.13
Max |XTE| (m)	1.97 ± 0.40	2.23 ± 0.68
Max |XTE| abs. (m)	3.78	—
Median |HE| (°)	2.75 ± 0.39	2.79 ± 0.57
Median speed (km/h)	7.12 ± 0.17	7.00 ± 0.29
Duration (s)	365 ± 52	383 ± 83

B. Baseline Navigation (Protocol 1)

Table I summarises the 10 baseline runs. The system achieves an **80% success rate** (8/10).

The 8 successful runs yield a median |XTE| of 0.53 ± 0.06 m (MAD), a median |HE| of $2.75 \pm 0.39°$, and a median speed of 7.12 ± 0.17 km/h. The maximum absolute XTE observed across all steps and all successful runs is **3.78 m** (run 5), illustrating the worst-case divergence the vehicle tolerates while still completing the route. The per-run XTE median ranges from 0.47 m (run 6, best) to 1.60 m (run 5, worst successful), a factor of 3.4× variability driven by the stochastic MB signal in the curve zone.

Fig. 5 shows the trajectories of all 10 runs overlaid on the learned route. The 8 successful trajectories closely follow the learned route, while the 2 failures (runs 4 and 8) diverge and stop before reaching the end.

Fig. 5: Baseline navigation: 10 runs overlaid on the learned route in CARLA Town05 (8/10 complete, median XTE $=$ 0.53 m, max absolute XTE $=$ 3.78 m). Solid lines: successful runs. Dashed lines with ×: failed runs. Run 4 stopped at WP 228; Run 8 stopped at WP 197.

TABLE II: Per-run baseline metrics (Protocol 1). XTE_{med} and XTE_{max} are computed over all steps of each run.

Run	$\|\text{XTE}\|_{\text{med}}$	$\|\text{XTE}\|_{\text{max}}$	$\|\text{HE}\|_{\text{med}}$	**Speed**	**Result**
0	0.66 m	2.34 m	2.8°	6.9 km/h	Success
1	0.55 m	1.37 m	2.5°	7.1 km/h	Success
2	0.99 m	2.39 m	3.6°	6.0 km/h	Success
3	0.48 m	1.42 m	2.8°	7.3 km/h	Success
4	2.71 m	3.83 m	6.8°	1.9 km/h	**Fail** (WP 228)
5	1.60 m	3.78 m	4.4°	3.7 km/h	Success
6	0.47 m	1.81 m	2.3°	7.2 km/h	Success
7	0.52 m	2.12 m	2.8°	7.1 km/h	Success
8	3.99 m	4.28 m	9.9°	0.1 km/h	**Fail** (WP 197)
9	0.49 m	1.68 m	2.2°	7.2 km/h	Success

The two failures (runs 4 and 8) share a common pattern: unfamiliarity progressively increases above the emergency threshold, speed drops to near zero, and the vehicle stops. Both fail after the curve zone, suggesting that a small tracking error accumulated during the curve can cascade into loss of recognition.

Table II provides per-run detail with both the median XTE and the per-run maximum, highlighting the worst-case divergence of each run. Runs 3, 6, and 9 are the most consistent ($\text{XTE}_{\text{med}} < 0.5$ m, $\text{XTE}_{\text{max}} < 2$ m). Run 5, despite a higher median (1.60 m) and a maximum of 3.78 m, completes the route, demonstrating that the model tolerates large instantaneous deviations as long as they do not persist.

Fig. 6 shows representative time-series data from run 0. The XTE oscillates within ± 1.5 m, the heading error stays below $\pm 15°$, speed adapts to the differential familiarity signal, and unfamiliarity for both MB circuits remains consistently low ($u < 0.05$ for most steps). Speed drops are correlated with high steering magnitude (curve zone, yellow band) and occasional unfamiliarity peaks.

C. Lateral Offset (Protocol 2)

Table III and Fig. 7 present the lateral offset results. The system succeeds at 8 out of 10 runs, including the extreme values of ± 2.0 m.

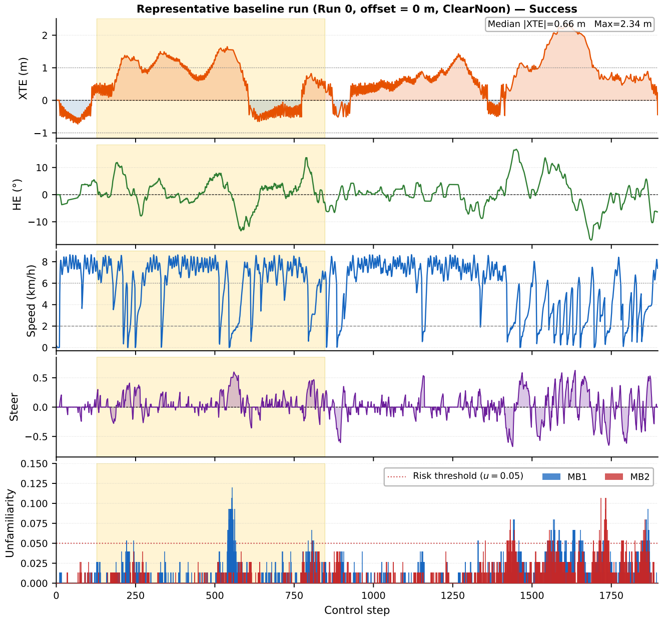


Fig. 6: Representative time-series of run 0 (baseline, offset = 0 m, ClearNoon — Success). From top to bottom: cross-track error XTE (orange: right of route, blue: left), heading error HE, vehicle speed (dashed lines: max 6 km/h and min 2 km/h), steering command, and unfamiliarity for both MB circuits. Yellow band: curve zone (WP 25–145). Speed drops correspond to sharp steering in the curve and transient unfamiliarity peaks.

TABLE III: Lateral offset results (1 run per offset, ClearNoon)

Offset	Med. $\lvert$XTE$\rvert$	Med. $\lvert$HE$\rvert$	Status
-2.0 m	0.82 m	3.6°	Success
-1.5 m	1.34 m	4.2°	Success
-1.0 m	3.31 m	17.0°	**Failure** (WP 187)
-0.5 m	3.19 m	3.8°	**Failure** (WP 190)
0.0 m (a)	0.87 m	3.6°	Success
0.0 m (b)	0.77 m	3.3°	Success
$+0.5$ m	0.56 m	3.1°	Success
$+1.0$ m	1.02 m	4.5°	Success
$+1.5$ m	0.74 m	2.9°	Success
$+2.0$ m	0.68 m	3.1°	Success

Two observations emerge. First, there is an asymmetry: both failures occur at negative offsets (-0.5 and -1.0 m). The route's main curve turns right, so a negative offset places the vehicle on the *outside* of the curve, where the required correction is stronger. Second, the fact that -1.5 m and -2.0 m succeed while -0.5 m and -1.0 m fail reveals a significant stochastic component—one run per offset is insufficient to draw definitive conclusions about the basin boundary.

For all successful runs, the final tracking performance converges to a median $|\text{XTE}| = 0.82 \pm 0.13$ m (MAD) regardless of initial offset, confirming the attractive nature of the learned route.

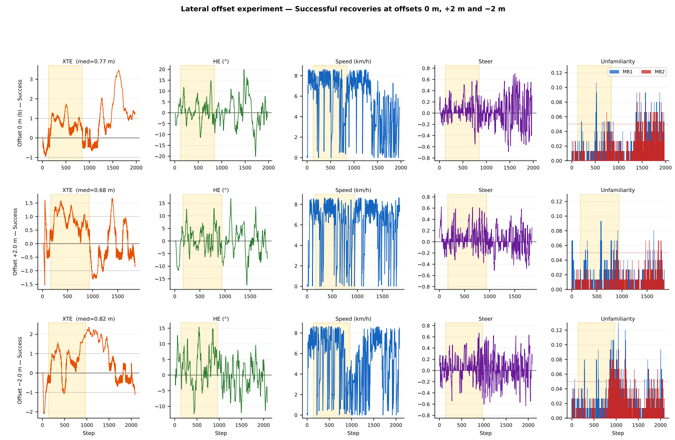


Fig. 7: Lateral offset experiment: time-series for three representative successful recoveries (offsets 0 m, $+2.0$ m, -2.0 m). Each row shows XTE, HE, speed, steering and unfamiliarity for one run. The XTE panel shows convergence toward the route centreline from the initial offset. Yellow band: curve zone. For the -2.0 m run, the negative XTE at startup reflects the left-side initial displacement.

D. Weather Sensitivity (Protocol 3)

The system fails under all four non-baseline conditions tested (Night, HardRain, Fog, Sunset). In each case, the vehicle stops within the first 50 waypoints due to the emergency brake threshold being exceeded. The failure mechanism is immediate: changed weather conditions (altered shadows, reflections, reduced contrast, or near-total darkness) produce Sobel gradient patterns that differ substantially from the learned memories, causing unfamiliarity to rise far above the operational threshold. Night conditions are the most severe, rendering the gradient patterns nearly unrecognisable. These results highlight the fundamental limitation of appearance-based encoding: the MB model stores raw gradient patterns rather than illumination-invariant features, making it highly sensitive to any change in lighting conditions. This behaviour mirrors that of real desert ants, which are reported to be disoriented when lighting changes between their outbound and return journeys [18].

VI. Discussion

A. Comparison with Real AntCar

Our results are consistent with the physical AntCar robot [10], which achieved 100% success (11/11 runs) on a 1.6 km real-world route with a median lateral error of 0.22 m. Our system yields 80% success with a median XTE of 0.53 m on a 300 m urban route. A direct numerical comparison is informative but must account for scale differences.

B. Scale-Normalised Comparison

A direct metric comparison between AntCar and our CARLA vehicle requires accounting for two scale factors: the physical size of the vehicle relative to the road, and the turning radius constraints.

TABLE IV: Model and navigation parameters

Parameter	Value	Origin
Visual resolution	32×32 px	AntCar [10]
Encoding method	Sobel gradient	AntCar
Kenyon Cells N_{KC}	15,000	AntCar
KC connections K	4	AntCar
Sparsity κ	0.005	Adapted
Oscillation range	$\pm 45°$, step $5°$	AntCar
Route length	300 m (300 WP)	This work
Waypoint spacing	1.0 m	This work
Control rate	$\sim$5 Hz	This work
Steer smoothing α	0.15	Tuned
Steer gain K_P	25	Tuned
Steer clip	± 0.75	Tuned
Max / min speed	6 / 2 km/h	Tuned
Stop threshold	$\lambda_{\min} < 0.75$	Tuned
MB random seeds	42, 99	Arbitrary

Lane-width normalisation. The AntCar robot (width $\approx$ 30 cm) operates in lanes of approximately 1.5 m, while the CARLA Tesla Model 3 (width $\approx$ 1.9 m) drives in Town05 urban lanes of approximately 3.5 m. Expressed as a fraction of lane width, the median XTE becomes 0.15 for AntCar versus 0.19 for our system—a difference of less than 5 percentage points, suggesting comparable lane-centring precision at scale.

Worst-case divergence. The comparison is less favourable for the maximum XTE: AntCar reaches at most $\approx$ 0.53 lane-widths, while our system reaches up to 1.08 lane-widths (3.78 m absolute maximum, run 5). This reflects two additional factors. First, the CARLA vehicle has a larger turning radius and heavier dynamics than a small robot, making recovery from large deviations slower. Second, our 5 Hz control rate (vs. $\sim$38 Hz for AntCar) reduces the responsiveness of the correction loop. These differences explain the higher worst-case divergence and the 20% failure rate, and motivate future work on faster control loops and improved steering dynamics.

Adaptation of κ. The adaptation to CARLA required reducing the sparsity parameter κ from 0.01 to 0.005 (see Section III). Additionally, the navigation control required significant retuning: the exponential smoothing ($\alpha = 0.15$), increased steer clip (± 0.75), and speed modulation in curves were all necessary to achieve stable navigation at the CARLA vehicle scale.

C. Negative Result: Parallel Learning

During the development phase, we investigated whether training both MB circuits simultaneously on the same set of augmented images (rather than using separate positive and negative oscillation sets) would simplify the pipeline. This parallel learning variant consistently produced degraded navigation performance, with both circuits converging to similar weight distributions and the differential signal $\Delta\lambda$ losing its directional sensitivity. The results were not reproducible across runs and no useful navigation was achieved. We therefore retained the original asymmetric augmentation strategy (positive oscillations for MB1, negative for MB2).

D. Biological Interpretation

The complete failure under changed weather is a *predicted consequence* of the biological model: desert ants are disoriented when lighting conditions change between outbound and return journeys [18]. The MB model encodes appearance-specific patterns, not illumination-invariant features. This is biologically faithful, as ant visual neurons respond to raw contrast, not abstract invariant representations.

The stochastic variability (20% failure rate in baseline, seemingly random lateral offset failures) mirrors the variability observed in real ant navigation [16]. The MB signal's quantization noise from the kWTA mechanism introduces inherent randomness, particularly in challenging segments like tight curves.

E. Limitations

Three main limitations emerge: (1) weather sensitivity makes the system impractical for real-world deployment without modifications; (2) tight curves remain a failure point due to the low-frequency MB signal; (3) only one run per condition in the offset and weather experiments limits statistical conclusions.

VII. Perspectives

Illumination-invariant encoding. Replacing Sobel gradients with robust descriptors (HOG, LBP) could reduce weather sensitivity while maintaining low computational cost. This would trade biological fidelity for practical robustness.

Multi-condition learning. Training under multiple weather presets would improve robustness, at the cost of biological fidelity. A compromise could involve learning under a small set of representative conditions (e.g., sunny and cloudy). However, the increased number of training patterns may saturate the KC layer, requiring careful tuning of κ.

Learning walks. Biologically-inspired learning walks [12], where the vehicle performs lateral and angular deviations during learning, could expand the attraction basin by providing the MB with training data under varied viewpoints.

Hybrid navigation. Combining the MB visual compass with path integration (odometry-based dead reckoning) could provide redundancy when the visual signal degrades, as real ants combine both modalities [18]. This would be particularly useful for navigating through transient occlusions or temporary appearance changes.

Dynamic environments. Testing with traffic and pedestrians in CARLA would assess robustness to dynamic scene changes—a practical requirement for urban deployment.

Multiple routes and scalability. The MB model's low memory footprint ($2 \times 15{,}000$ weights $= 120$ kB) suggests that many routes could be stored simultaneously. Implementing a route selection mechanism would demonstrate the scalability of the approach.

VIII. Conclusion

We implemented ant-inspired visual navigation using the Mushroom Body neural model in the CARLA Town05 urban simulator. The system achieves 80% success on a 300 m route with a median cross-track error of 0.53 m (maximum 3.78 m), and lane-normalised precision comparable to the original AntCar robot. The attraction basin extends to ± 2 m, confirming the model's ability to converge toward the learned route. However, systematic failure under all altered weather conditions highlights the fundamental limitation of appearance-specific encoding—a property shared with the biological system that inspired this work. Future work should explore illumination-invariant visual features and multi-modal integration to bridge the gap between biological fidelity and practical robustness.

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